

Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 5J nr. 2.1

$$L = \int_0^{2\pi} \sqrt{(x'(t))^2 + (y'(t))^2} dt$$

$$x'(t) = a(1 - \cos(t))$$

$$y'(t) = a(0 + \sin(t)) = a \sin(t)$$

$$\Rightarrow \int_0^{2\pi} a \sqrt{(1 - \cos(t))^2 + (\sin(t))^2} dt$$

$$= \int_0^{2\pi} a \sqrt{1 - 2\cos(t) + (\cos^2(t) + \sin^2(t))} dt$$

$$= \int_0^{2\pi} a \sqrt{1 - 2\cos(t)} dt$$

$$= \int_0^{2\pi} a \sqrt{2(1 - \cos(t))} dt$$

$$= a\sqrt{2} \int_0^{2\pi} \sqrt{1 - \cos(t)} dt$$

verdubbeling hoek: $\cos(t) = 1 - 2\sin^2\left(\frac{t}{2}\right)$

$$= a\sqrt{2} \int_0^{2\pi} \sqrt{1 - 1 + 2\sin^2 \frac{t}{2}} dt$$

$$= a\sqrt{2} \int_0^{2\pi} \sqrt{2\sin^2 \frac{t}{2}} dt$$

$$= a\sqrt{2} \int_0^{2\pi} \sqrt{2} \sin \frac{t}{2} dt$$

$$= 2a \int_0^{2\pi} \sin \frac{t}{2} dt$$

$$\begin{aligned}
 &= 4a \left[-\cos \left(\frac{t}{2} \right) \right]_0^{2\pi} \\
 &= 4a(-\cos \pi + \cos 0) = 8a
 \end{aligned}$$

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References